An Introduction

Database:

An organized collection of Data;

An organized collection of records.

Data:

Unit of information/record is the raw facts that forms information.

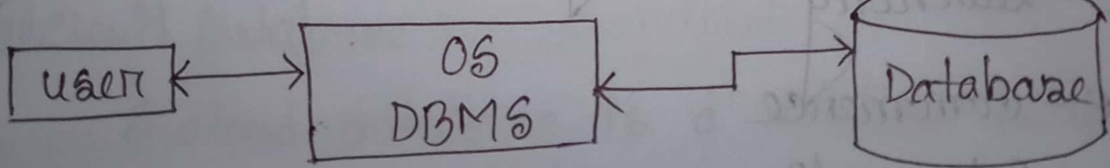
Information/record/entity:

Some meaningful and organized collection of data.

DBMS:

Is a computerized system that is liable or responsible for defining, creating, manipulating processing, accessing and using the database.

Oracle,



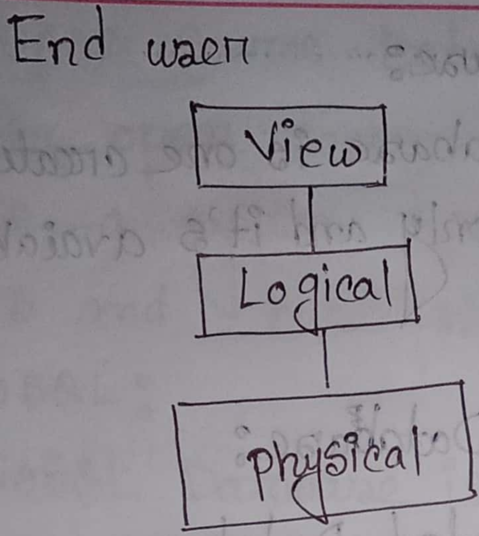
Components (buildings, functional)

Word document	Database
Letter	Character (a-z, A-Z, 0-9, sp-ch)
Word	Field / attribute / column / string (ID, name)
Sentence	Record / Entity
Paragraph	Entity / set / records
Essay / Document	Database

Applications:

Banking
Hospital
parlour
Mobile communication
Ticketing
student record management (school, college, varsity
e-learning
e-commerce
social media

Data Abstraction:



Types of Database:

(I) centralized Database:

A centralized database is basically a type of database that is stored, located as well as maintained at a single location only.

(II) Distributed Database:

A distributed Database management system is a set of multiple, logically interrelated database distributed over a network.

(III) Cloud Database:

A cloud database is a database built to run in a public or hybrid cloud environment to help organize, store and manage data within an organization.

(iv) Commersial Datadase:

A Commercial Database is one created for commercial purposes only and it's avaiable at a price.

(v) Object Oriented Database:

An object Oriented Database management system is a database management system is a database management system (DBMS) that supports the modelling and creation of data as objects.

(vi) Temporal Database:

A temporal database management system stores data relating to time instances.

(vii) Spatial Database:

Spatial database organize data for inventorying and querying databases, conducting spatial analysis and creating map visualizations within an integrated manner for managing large data stores.

(VIII) Open Source Database:

An open source database allows users to create a system based on their unique requirements and business needs.

(IX) NoSQL:

NoSQL Database is used to refer a non-SQL or non relational database.

(X) Graph Database:

Graph databases are purpose-built to store and navigate relationships

(XI) Network Database:

A network database management system is based on a network data model, which allows each record to be related to multiple primary records and multiple secondary records.

(XII) Document Database:

A document database is a type of nonrelational database that is designed to store and query data as JSON-like documents.

File system vs Database system :

1) Data redundancy (redundant) and inconsistency - it maintains data consistency.

2) Data Isolation - No problem in data isolation

3) Difficulties in data access - Easy data access.

4) Concurrent access anomalies - allowing concurrent access.

5) Atomicity problem - Allowing atomicity

6) Security problem - No security problem

Database Language :

SQL - Structured query language (sequel)

DDL - Data Definition language (Alter, creat)

DML - Data Manipulation language (select, delete)

DCL - Data Control language (grand, revoke)

TCL - Transaction control Language.

Database users:

a. Naive user - end user

b. Application programmer

Rapid application development (RAD) - visual basic, Javascript, HTML

API - Application programming interface

ODBC - Open database connectivity

JDBC - Java database connectivity

c. Sophisticated user

OLAP - Online analytical processing tool

d. Specialized user

Expert system - KB, DB

A (DBA) -> B (DBMS) -> C (data and program)

DBA Functions:

1. Schema Definition
2. Access method definition
3. Physical Organization modification

4. Granting of authorization of data access

5. Routine Maintenance

Functional Components of database:

1. Query processor:

   a) DDL interpreter

   b) DML Compiler

   c) Query Evaluation Engine

2. Storage manager

   a) Authorization & integrity manager

   b) Transaction manager

   c) File manager

   d) Buffer manager

Data Abstraction:

The process or mechanism of hiding data

1) View level:

This is the highest level of abstraction. Only a part of the actual database is viewed by the users. This level exists to ease the accessibility of the database by an individual user.

2) Logical Level:

The logical or conceptual level is the intermediate or next level of data abstraction. It explains what data is going to be stored in the database and what the relationship is between them. It describes the structure of the entire data in the form of tables

3) physical level:

This is the layer of data abstraction where the raw data is physically stored as files. This layer contains all the complex data structures and the data accessing methods defined. The physical layer is the lowest level of data abstraction is a DBMS.

Schema:

A database schema defines how data is organized within a relational database; this is inclusive of logical constraints such as, table names, fields, data types and there relationship between these entities.

Instance:

A database instance is a set of memory structures that manage database fiels. A database is a set of physical fiels on disk created by the CREATE DATABASE statement.

Database language:

Database language is 3 types.

① DDL :

Data definition Language.

DDL Definition Language (DDL) is a subset of SQL. It is a language for describing data and its relationships in a database.

(ii) DML :

Data Manipulation Language.

Data is ~~an~~ represents a collection of programming language explicitly used to make changes to the database such as: CRUD operations to creat, read, update and delete data.
Using INSERT, SELECT, UPDATE and DELET commands.

DCL :

Data Control Language.

DCL statements are primarily used to grant or revoke privilages to database.
→ Procedural Language : Everything is defined
→ Non- procedural Language

Database Administrator (DBA)

Who has complete control over the database.

Function of database Administrator :

(i) Schema definition
(ii) Storage structure and access method definition.

(iii) Schema and physical organization modification

(iv) Grrating of authorization.

(v) Periodic Backup

- Data Backup
- Data clean
- Performance

Functional components of Database:

Query processor

- DDL interpreter — which interprets DDL statements and records the definitions in the data dictionary
- DML compiler — again as the name suggests compiles the DML statements such as select, update & delete into low-level instructions.
- Query Evaluation Engine — which excutes low-level instructions generated by the DML compiler

Storage Manager

- Integrity & Authorization Manage
- Transaction Manager
- File Manager
- Buffer Manage

Addition data structure

- Data files — which store the database itself
- Andicies — which provide fast access to data items that hold particular values
- Data Dictionary — which store metadata about the structure of the database, in particular the schema of the database

User
↓
Query tools/API
↓
Query Processor
↓
Storage Manager
↓
Database

* SQL এর Mathematical form Relational Algebra.

Data Model:

A database Model defines the logical design and structure of a database and defines how data will be stored, accssed and updates in database management system.

4 Types of Data Model:

① Hierarchical Model: it is a data model in which the data are organized into a tree-like structure.

(ii) Network Model: a hierarchical Model that is used to represent the many-to-many relationship among the database constraints.

(iii) Entity-Relationship Model: is used to define the data elements and relationship for a specified system.

(iv) Relational Model: an approach to logically represent and manage the data store in a database by storing data in tables.

CamScanner

Types of Relational Model:

(i) One-to-One:

a relational between an instance of an entity with another.

(ii) One-to-Many:

When one row in table A is linked to many rows in table B, but only one row in table B is linked to one row in table A.

(iii) Many-to-Many:

When one or more items in one table can have a relationship to one or more items in another table.